

Manufacturing Guide

Passive Temperature Compensation Device (PTCD)

Noble Aggie Labs

Revision 1.0

June 2026

CRITICAL SAFETY NOTICE

Manufacture of this device involves **sharp tools**, **high temperatures**, and dangerous assembly processes such as **electron-beam welding**. Failure to follow these instructions may result in **serious injury or death**.

- All personnel must be trained in their shop's applicable safety protocols before beginning any operation in this guide.
- **Machining:** Sharp cutting tools cause severe lacerations. Never reach into a spinning workpiece or move your hands near an unguarded cutter. Wear safety glasses at all times on the shop floor. Flying chips are a serious eye hazard.
- **Band saw:** Keep fingers clear of the blade at all times. Use a "chicken finger" to manipulate small stock. Do not force any cut.
- **Heat treatment:** The annealing furnace operates at temperatures up to 1040 °C. Wear heat-resistant gloves and a face shield when loading, unloading, or working near an open furnace. Allow parts to cool for at least three hours before handling.
- **Electron-Beam Welding (EBW):** EBW generates high-voltage X-rays and an intense electron beam. Only trained and certified EBW operators may perform welding operations. The operator must confirm the vacuum chamber is sealed and all radiation interlocks are engaged before initiating any weld.

- **Electrical Discharge Machining (EDM):** Dielectric fluid is a skin and eye irritant and a fire hazard. Wear chemical-resistant gloves. Ensure the machine is fully de-energised before accessing the work zone. Properly dispose of spent dielectric fluid.
- **General:** Do not work alone on any operation. Wear appropriate personal protective equipment (PPE) at all times, including safety glasses, closed-toe shoes, and long pants. Do not wear loose clothing or jewellery while in the shop.

1 Introduction and Scope

This guide details the complete manufacture, assembly, and calibration of the Passive Temperature Compensation Device (PTCD), a thermomechanical valve that passively adjusts a microscopic orifice in response to ambient air temperature.

1.1 Required Tools

The following tools and machines are required:

- Computer-Numeric Control (CNC) mill and lathe that can meet or exceed the dimensional tolerances in Appendix ??
- Band saw
- Annealing/heat-treatment furnace capable of reaching 1040 °C
- Sinker EDM (Electrical Discharge Machining) capability
- Electron-Beam Welding (EBW) capability
- Two perpendicular granite surfaces. One surface must have a drilled hole whose center is 20 mm from the intersection between the two surfaces. The hole must measure at least 3.4 mm in diameter and 5.1 mm depth.
- New, carbide-steel tools. Note: this is important to properly machine the parts made of Invar 36, which work-hardens as it is cut. **[TODO: Specify which bits are needed]**
- T6 hand screwdriver

- Loctite 243 or similar permanent thread sealant
- Cutting fluid for all operations

1.2 Coordinate System

Coordinate convention used throughout this guide:

- **X (lateral):** points to the right as one looks downstream along the gas flow.
- **Y (vertical):** points directly upward.
- **Z (longitudinal):** points upstream along the direction of gas flow.

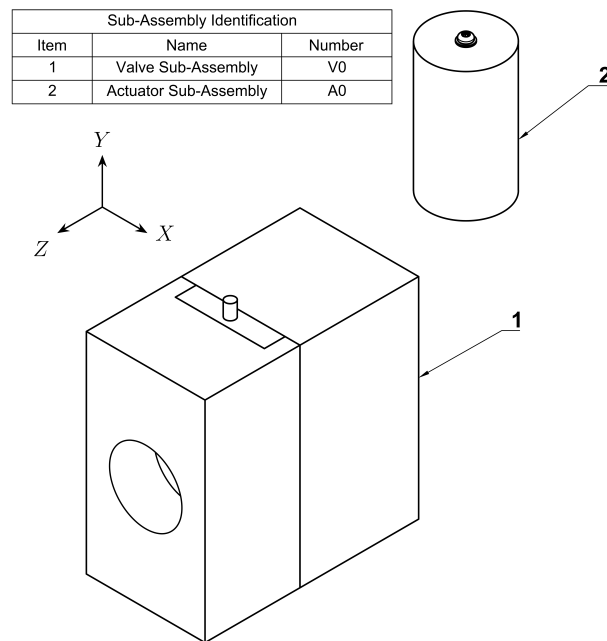


Figure 1: *Reference axes*

2 Part Identification

Before beginning, familiarise yourself with every part using the figures below and the Bill of Materials in Appendix ??.

Parts List				
Item	Qty	Part Name	Part Number	Material
1	1	Housing Inlet v6	V2	Invar 36
2	2	Housing Center v3	V1	Invar 36
3	1	Housing Outlet v4	V3	Invar 36
4	1	Valve Orifice v3	V4	Invar 36
5	2	Bellows v2	I718-159-72-430	Inconel 718

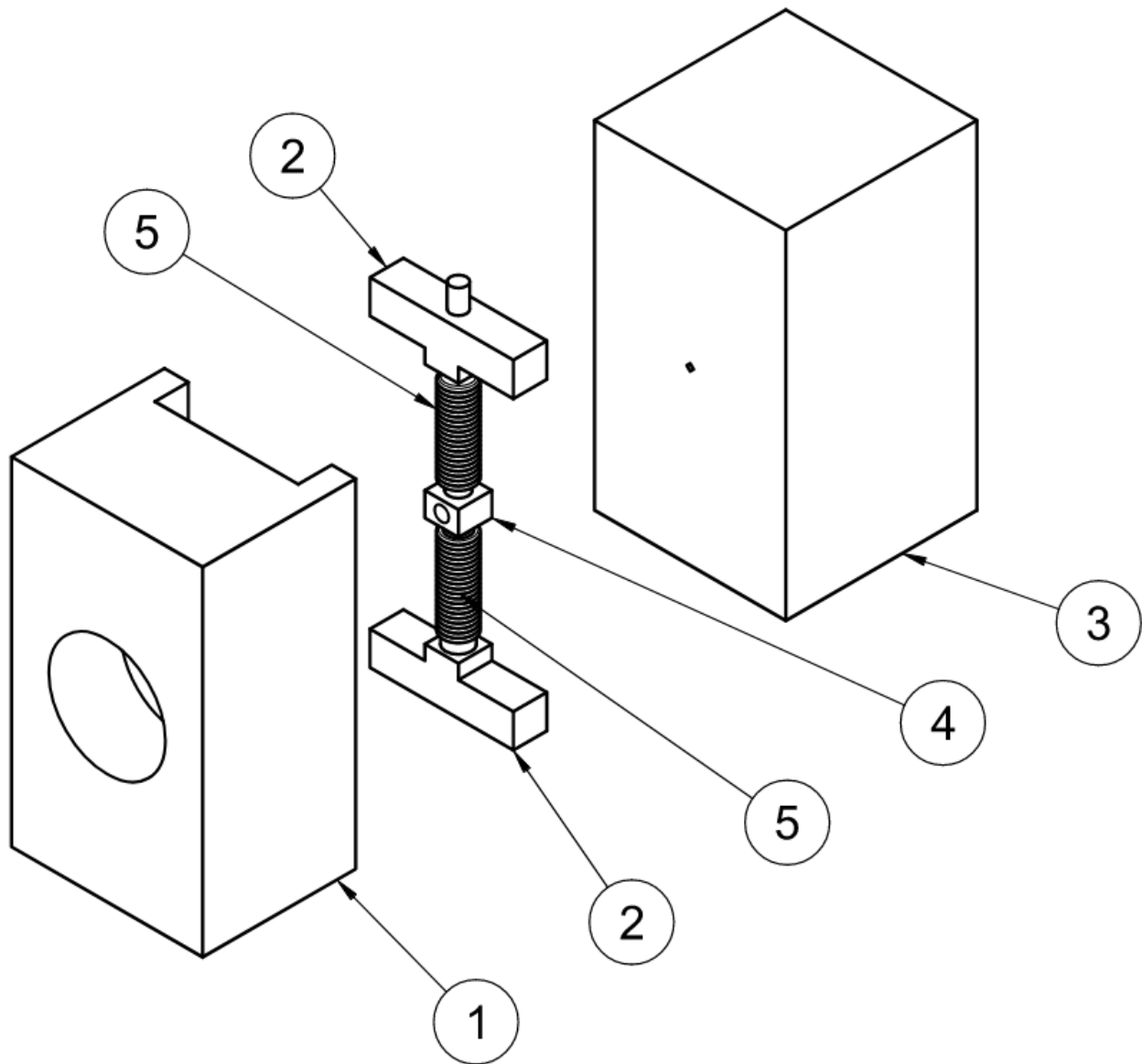


Figure 2: *Valve subsystem part identification*

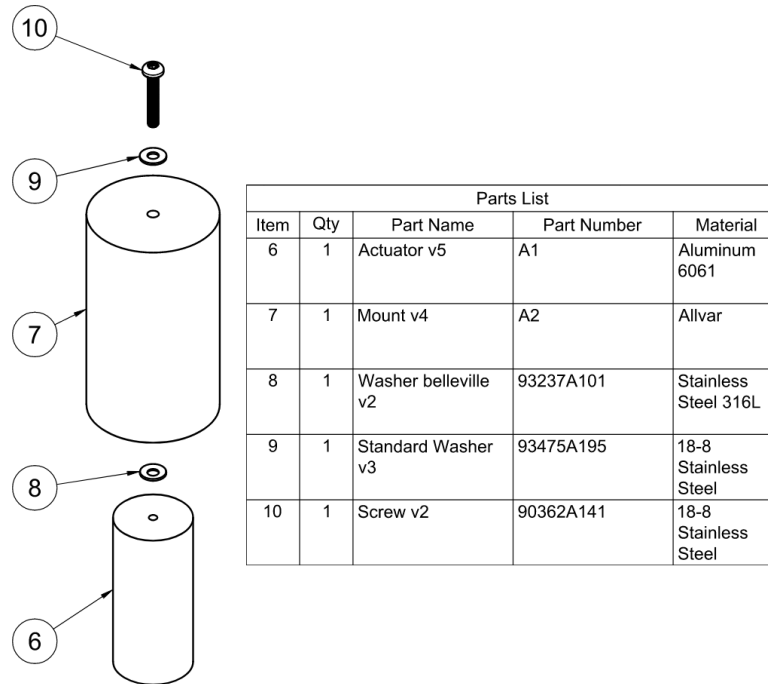


Figure 3: *Actuator subsystem part identification*

Table 1 summarises all parts manufactured or purchased for this device.

Table 1: *PTCD Bill of Materials summary.*

Part No.	Name	Qty	Material	Source
V1	Housing Center	2	Invar 36	Machined from billet stock
V2	Housing Inlet	1	Invar 36	Machined from billet stock
V3	Housing Outlet	1	Invar 36	Machined from billet stock
V4	Valve Orifice	1	Invar 36	Machined from billet stock
I718-159-72-430	Bellows	2	Inconel 718	Purchased (Mini-Flex)
A1	Actuator	1	Al 6061-T6	Machined from rod stock
A2	Mount	1	ALLVAR 30	Machined from rod stock
93237A101	Belleville Washer	1	316 SS	Purchased (McMaster-Carr)
93475A195	Standard Washer	1	18-8 SS	Purchased (McMaster-Carr)
90362A141	Screw M2×0.4×12	1	18-8 SS	Purchased (McMaster-Carr)

3 Manufacturing

3.1 General Machining Notes

Invar 36 work-hardens during machining. Use **new carbide tooling** throughout. Replace tooling at the first sign of wear; worn tools exacerbate work hardening and degrade surface finish on critical faces.

Annealing schedule for all Invar 36 parts:

1. **Before machining:** anneal at 1040 °C (1900 °F) for 60 minutes to reduce as-received hardness and improve machinability [? , p. 6].
2. **After all machining is complete:** anneal at 93 °C (200 °F) for 48 hours to relieve residual stress and improve long-term dimensional stability [? , p. 6].

Allow parts to air-cool to room temperature before measuring or proceeding to the next operation.

Referencing convention: Before beginning each machining operation, identify which face will serve as the primary datum. As indicated in the drawings, each of these faces must be carefully finished to minimize dimensional errors for critical features.

[TODO: Bit sizes and orientations for all operations]

3.2 Invar 36 Billet Stock: V1, V2, V3, V4

Starting stock: 44.45 mm × 76.2 mm × 82.55 mm (1 3/4" × 3" × 3 1/4") Invar 36 billet. These dimensions will be oriented along the **lateral (X)**, **vertical (Y)**, and **longitudinal (Z)** dimensions, respectively.

WARNING — Band Saw Hazard Keep fingers and hands clear of the band saw blade at all times. Use the fence or a chicken finger to guide stock. Never reach over or behind the blade during a cut. Allow the blade to come to a complete stop before adjusting the workpiece.

Step 1 — Cross-cut longitudinal axis into blanks

Using a metal-cutting band saw, cross-cut the longitudinal axis (Z) to produce blanks of the following *minimum* lengths, leaving additional length for subsequent facing operations:

- Housing Inlet (V2): 32 mm
- Housing Outlet (V3): 40 mm
- Housing Center (V1) / Valve Orifice (V4): 7 mm

Account for the band saw blade kerf thickness when planning cut locations. Typical metal band saw kerfs are 1 mm to 2 mm wide. Ensure adequate additional material remains on each blank after accounting for kerf and facing stock.

Step 2 — Cross-cut lateral axis to separate V1 and V4 slugs

Cross-cut the 7 mm slug along the **lateral axis (X)** to produce:

- Housing Center $\times 2$ (V1): 10 mm lateral length each
- Valve Orifice (V4): 7 mm lateral length

You now have five rough blanks: V2, V3, two pieces of V1, and V4.

Step 3 — Anneal blanks

To soften the material before machining the part features, place all five blanks in the annealing furnace. Anneal at 1040 °C for 60 minutes. Allow to air-cool to room temperature before proceeding.

Housing Center (V1) — $\times 2$

Refer to drawing V1. Produce both Housing Centers identically.

1. **Mill all six faces** to bring the part to nominal rectangular dimensions.
2. **Mill two T-slots on the lateral sides** (+X and −X faces) for the valve orifice sliding interface.
3. **Drill through-hole** for the valve stem passage. Use a reamer to bring to final H7 tolerance.

Housing Inlet (V2)

Refer to drawing V2.

1. **Mill all six faces** to nominal rectangular dimensions.

2. **Finishing pass on critical faces** (the +X and +Y faces).

The +X face is the primary reference for the Housing Inlet. Machine this face last among the milled faces, with a dedicated finishing pass using a sharp tool at a slow feed rate and light depth of cut. The resulting surface must be mirror-smooth; verify with a surface-finish gauge or profilometer. This face will be used for critical dimension measurements and will be held flush with the corresponding face on the Housing Outlet during EBW assembly.

3. **Mill top T-groove** on the +Y face for the upper valve stem channel.
4. **Mill bottom T-groove** on the -Y face for the lower valve stem channel.
5. **Mill central channel** through the +Z face to form the sliding pocket for the Valve Orifice.
6. **Drill large central hole** through the inlet channel. This is the main gas passage.
7. **Drill small inlet hole** for the upstream pipe connection.
8. **Tap large hole** with G 3/4-14 A (in) thread (BSP pipe thread) for the upstream pipe fitting. Use a pipe tap and cutting fluid.

Housing Outlet (V3)

Refer to drawing V3.

1. **Mill all six faces** to nominal dimensions.
2. **Finishing pass on Critical Faces A, B, and C** (the +X, +Y, and +Z faces respectively).

The +X and +Y faces are the primary references for the Housing Outlet. Machine these faces last among the milled faces, with a dedicated finishing pass using a sharp tool at a slow feed rate and light depth of cut. The resulting surfaces must be mirror-smooth; verify with a surface-finish gauge or profilometer. These faces will be used for critical dimension measurements and will be held flush with the corresponding faces on the Housing Inlet during EBW assembly.

The +Z face must be similarly finished to ensure even abutment against the Housing Inlet.

3. **Drill large central hole** as in the Housing Inlet.
4. **Plunge small orifice hole** with an end mill.
5. **Tap large hole** with G 3/4-14 A (in) thread for the downstream pipe fitting.
6. **Sinker EDM — square through-orifice:**

WARNING — EDM Hazard The dielectric fluid used in EDM is a fire hazard and a skin/eye irritant. Wear chemical-resistant gloves and eye protection. Ensure the EDM machine is fully de-energised before accessing the work zone. Dispose of spent fluid per facility hazardous-waste procedures.

Machine the square through-orifice using sinker EDM. Critical requirements:

- The **upper interior corner** of the square orifice must have as small a fillet radius as possible. This is the mission-critical corner; a larger fillet directly degrades the effective diameter accuracy.
- Minimise EDM recast (white) layer on all orifice edges; no micro-cracking is permitted.
- Thoroughly clean the part after EDM to remove all dielectric fluid and microscopic debris before proceeding.
- The horizontal and vertical position of the critical upper corner must be held as exactly as possible. Verify with optical metrology after EDM.

EDM Integrity Verification

Confirmation of the above requirements requires a destructive test. Perform an identical operation to the one specified on a sample workpiece. Carefully cut the piece in two along its central axis and optically verify that no EDM recast or micro-cracking is present.

Valve Orifice (V4)

Refer to drawing V4.

1. **Mill $\pm Z$ and $\pm X$ faces** to bring part to nominal rectangular dimensions.
2. **Finish all faces** except for the -Y face, paying especial attention to the +X face.
3. **Plunge small hole** with an end mill.

4. **Sinker EDM — square through-orifice (leading orifice):** Machine the square through-orifice using sinker EDM as per the above EDM operation.
5. **TiN PVD coating:** Apply a TiN (titanium nitride) Physical Vapor Deposition coating to the four contact surfaces per drawing V4, Note 5. Confirm the coating thickness does not exceed 0.004mm; re-measure the g6 dimensions after coating and verify the fit is maintained before proceeding to assembly.

3.3 6061 Aluminum Rod Stock: A1 (Actuator)

Starting stock: $\varnothing 19.05\text{ mm} \times 76.2\text{ mm}$ ($\varnothing 3/4'' \times 6''$) 6061-T6 Aluminum rod.

1. **Band saw — cut to length:** Cross-cut the rod to a length slightly above nominal; leave material for facing.
2. **Mill to length:** Face both +Y and −Y ends on the milling machine to remove band saw marks and bring the part close to nominal length.
3. **Finishing pass on +Y face (top):**

The length of the actuator is a **critical dimension**: it directly sets the actuator expansion rate.

4. **Lathe — turn outside diameter:** Turn the outer diameter.
5. **Drill central hole:** Drill the axial hole through the top of the part, centred on the axis.
6. **Tap:** Tap M2×0.4 6H thread in the central hole for the calibration screw.

3.4 ALLVAR 30 Rod Stock: A2 (Mount)

Starting stock: $\varnothing 25.4\text{ mm} \times 45.72\text{ mm}$ ($\varnothing 1'' \times 1.8''$) ALLVAR Alloy 30 rod.

1. **Band saw — cut to length:** Cross-cut the rod slightly above nominal.
2. **Mill to length:** Face both ends to remove saw marks and bring to near-nominal height.
3. **Finishing pass on −Y face (bottom):**

The $-Y$ face is electron-beam welded to the top of the housing. A flat, clean surface is essential for the intimate joint required by EBW. Machine this face last with a light finishing pass. Verify flatness with a surface plate and feeler gauge before proceeding.

4. **Drill small clearance hole:** Drill the center through hole to a minimum depth of 5 mm.
5. **Drill large inner hole:** Drill the bulk of the large hole. Leave ample diametral and axial clearance for precise dimensions.
6. **Bore large inner hole:** Carefully bore the hole's inner diameter to tolerance. Do not bore the entire hole; leave axial clearance at the end to avoid damaging the end face.
7. **Plunge large inner hole:** Plunge the hole with a long end mill. Perform a finishing pass to bring the hole's length to tolerance and to square the circumference of its inner face.
8. **Turn outer diameter:** turn the outer diameter with a lathe.

4 Assembly

WARNING — Electron-Beam Welding (EBW) Hazard

EBW generates high-voltage X-ray radiation and a focused electron beam. **Only trained, certified EBW operators** may perform these operations. Confirm all radiation interlocks are active and the vacuum chamber is sealed before initiating any weld. No EBW operation may be performed by untrained personnel under any circumstances.

4.1 General Assembly Notes

All permanent joints in the PTCD, except the actuator-mount connection secured by the calibration screw, are made by Electron-Beam Welding (EBW). EBW requirements (drawing W1, W2 notes) apply to all weld operations:

1. Welding process: EBW (autogenous; no filler metal).
2. Mating surfaces must be intimate; maximum gap ≤ 0.01 mm.

3. All surfaces to be welded must be thoroughly cleaned and free of oil, grease, and oxidation prior to welding. Use appropriate solvent (e.g. isopropyl alcohol) and clean-room wipes.
4. Purge internal cavities with inert gas (e.g. argon) during welding to prevent oxidation and spatter.
5. All wetted EBW joints must be subjected to 100% X-ray radiographic examination (RT) to confirm a weld joint efficiency $E = 1.0$ per ASME B31.3 [?].

EBW requires an unobstructed line of sight to the weld joint throughout the full rotation of the workpiece. Plan each setup carefully. The sub-assembly sequence in Sections 4.2 and 4.3 is designed to maintain 360° weld access at every stage.

4.2 Valve Sub-Assembly

Required tooling for valve assembly:

- Two perpendicular granite surface plates. One plate must contain a clearance hole at least 3.4mm in diameter and at least 5 mm deep at precisely 20 mm from the intersection between the plates (to accept the lower valve stem during step 4.2).
- Clean lint-free gloves and wipes.
- Optical comparator or profilometer for verifying face flush and orifice position.

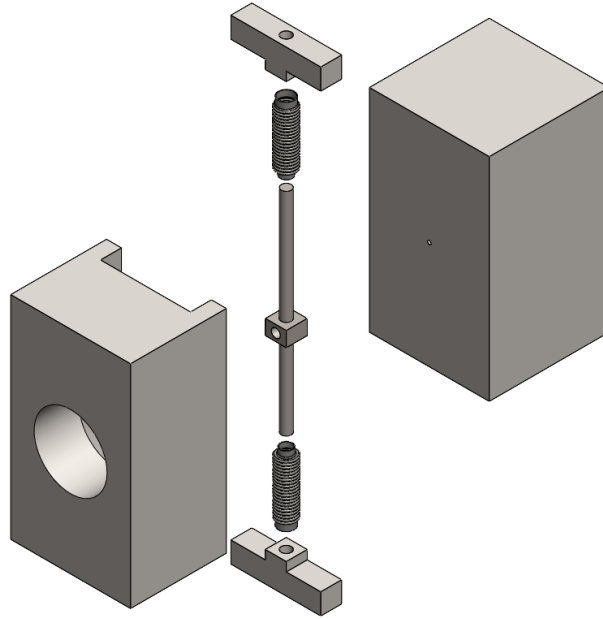


Figure 4: *Valve subsystem exploded view*

Step 1 — Weld bellows to Valve Orifice (V4)

1. Orient the first (upper) bellows with its **small neck (Neck 1) toward the center** of the Valve Orifice. Seat it on the upper valve stem stub of V4.
2. Orient the second (lower) bellows identically: small neck toward center, seated on the lower valve stem stub.
3. EBW both bellows to V4 with circumferential welds.

Step 2 — Add Housing Centers (V1) to valve stems

1. Slide one Housing Center (V1) onto the upper valve stem of the V4–bellows sub-assembly through its bore. The V1 piece should slide freely without binding.
2. Slide the second Housing Center onto the lower valve stem.

Step 3 — Lay sub-assembly on granite surface and tack-weld

1. Set the sub-assembly on the granite surface plate, with the -Z face against the surface.
2. Apply **two spot welds** at each bellows–Housing Center connection to fix their position.

Step 4 — Full weld of Housing Centers to bellows

1. Remove the sub-assembly from the granite surface.
2. Fully EBW each Housing Center to its bellows with a circumferential weld.

Step 5 — Insert valve sub-assembly into Housing Inlet (V2)

1. Verify that the V4 coated surfaces are clean and the housing channel in V2 is free of debris.
2. Slide the sub-assembly (V4 + bellows + V1 $\times$ 2) into the central channel of the Housing Inlet (V2).

Step 6 — EBW Housing Centers to Housing Inlet; verify face flush

1. With the sub-assembly fully seated in V2, EBW the Housing Centers (V1) to the Housing Inlet (V2).

Step 7 — Finish the Housing Inlet face

1. Make a finishing pass across the whole -Z face of the subassembly to ensure flush abutment against the Housing Outlet. Be careful not to damage the bellows while cutting.

Step 8 — Connect the Housing Outlet

1. Invert the assembly and set its +Y and +X faces flush against the two granite surfaces. Use the provided hole to make room for the protruding valve stem.
2. Set the Housing Inlet in like manner against the granite surfaces. Butt its +Z face against the assembly's -Z face.
3. Weld the assembly along its exposed edges enough to secure the parts in place. Remove from the granite surfaces; fully weld the connection.

Step 9 — Finish the Upper Surface

1. Perform a finishing pass on the +Y surface of the assembly. Maintain clearance around the valve stem at a minimum of 1 mm to avoid damage and a maximum of 7 mm to ensure the full mount-housing contact surface is finished.

4.3 Actuator Sub-Assembly



Figure 5: *Actuator subsystem exploded view*

Step 1 — EBW Actuator (A1) to upper valve stem

1. EBW the actuator to the valve stem. Confirm the actuator and valve stem are axially aligned; any tilt introduces an off-axis load on the orifice.

Thermal equilibration

Before proceeding, allow all parts to thermally equilibrate with their environment for a full 60 minutes. This ensures that the device can be calibrated before the thread lubricant sets.

Step 2 — Add Mount and Calibration Assembly

1. Apply Loctite 243 or a similar lubricant-sealant to the screw.
2. Invert the mount (A2). Insert the screw (90362A141) through the standard flat washer (93475A195) and mount. Slide the Belleville washer (93237A101) onto the screw, conical side facing toward the mount top when righted.
3. Invert the housing-actuator assembly. Carefully insert the actuator into the the mount until it hits the screw. Thread the screw into the actuator until the mount can sit flush against the housing and the Belleville washer is slightly compressed.

4. Right the assembly and EBW the mount to the housing.

Time-sensitive calibration Perform the calibration steps detailed in Section 5 immediately to ensure the thread sealant does not set.

5 Calibration

Calibrate the PTCD at approximately $T_{ref} = 25\text{ }^{\circ}\text{C}$ (room temperature).

Calibration Procedure

1. **Thread the gas supply.** Connect the PTCD to a controlled gas flow supply via its threaded interfaces. Precisely measure the ambient temperature and the pressure differential across the device.
2. **Calculate required pressure drop.** Use the following equations to determine the required pressure differential for the current temperature:

$$D_{\text{effective}} = A_1(T_{\text{ambient}} - T_{\text{reference}}) + A_0 \quad (1)$$

$$A_1 = -1.524 \cdot 10^{-3} \frac{\text{mm}}{^{\circ}\text{C}} \pm 5\% \quad (2)$$

$$T_{\text{reference}} = 25\text{ }^{\circ}\text{C} \quad (3)$$

$$A_0 = 0.254\text{ mm} \pm 5\% \quad (4)$$

$$A = \frac{\pi}{4} D_{\text{effective}} \quad (5)$$

$$Q = C_d A \sqrt{\frac{2\Delta P}{\rho(1 - \beta^4)}} \quad (6)$$

Refer to White's Fluid Mechanics[? , eq. 6-102] for interpretation of equation (6).

3. **Adjust the orifice.** Using a T6 Torx driver, turn the calibration screw in small increments. Turn it clockwise to increase the effective diameter and decrease the pressure differential.
4. **Depressurise and disconnect.** Let the system run at steady-state for six hours to allow the sealant to fully set under these conditions. Then close the supply valve and vent the downstream line before disconnecting any fittings.